Frontier



University of Alberta

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

University of Alberta

Organization location (we welcome applicants from anywhere in the world)

Edmonton, Alberta, Canada

Name(s) of primary point(s) of contact for this application

Sasha Wilson

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

University

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Optimizing acid heap leaching of serpentinite for enhanced mineralization with critical metal recovery

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

We are developing a scalable carbon dioxide removal (CDR) approach that accelerates carbonation of ultramafic (magnesium-rich) mine tailings through acid heap leaching, while co-recovering critical metals such as nickel and cobalt. This process integrates three key components: (1) enhanced mineral dissolution, (2) acid/base generation with brine recycling and (3) mineral carbonation (**Figure 1**).

Our technology will advance **Frontier's Enhancing Surficial Mineralization** priority area by targeting the key bottleneck in mineral-based CDR: the slow release of divalent cations (e.g., Mg^{2+}) from abundant alkaline earth silicate minerals. We have shown that this barrier can be overcome using heap leaching to rapidly liberate Mg while simultaneously enhancing Ni and Co recovery, creating an additional revenue stream that may partially or fully offset CDR costs. By coupling accelerated mineral dissolution with critical metal recovery, our approach embeds resource extraction into enhanced surficial mineralization, offering a pathway toward step-change gains in both scale and cost. An additional benefit of this process is the destruction of chrysotile asbestos fibres at suitable mines (e.g., Thetford, Québec, Canada).

In contrast to conventional surficial enhanced weathering approaches that rely on comparatively slow, passive natural reactions and/or are heavily constrained by the abundance of brucite [$Mg(OH)_2$], our approach achieves order-of-magnitude increases in reaction kinetics (Hamilton et al., 2020; Turvey et al., in review). Crucially, this approach can also integrate bioleaching, using acid-generating microbes to reduce dependence on externally supplied reagents (Power et al., 2010), which is one of the greatest costs and scalability constraints identified in technoeconomic assessments (Su et al., in review).

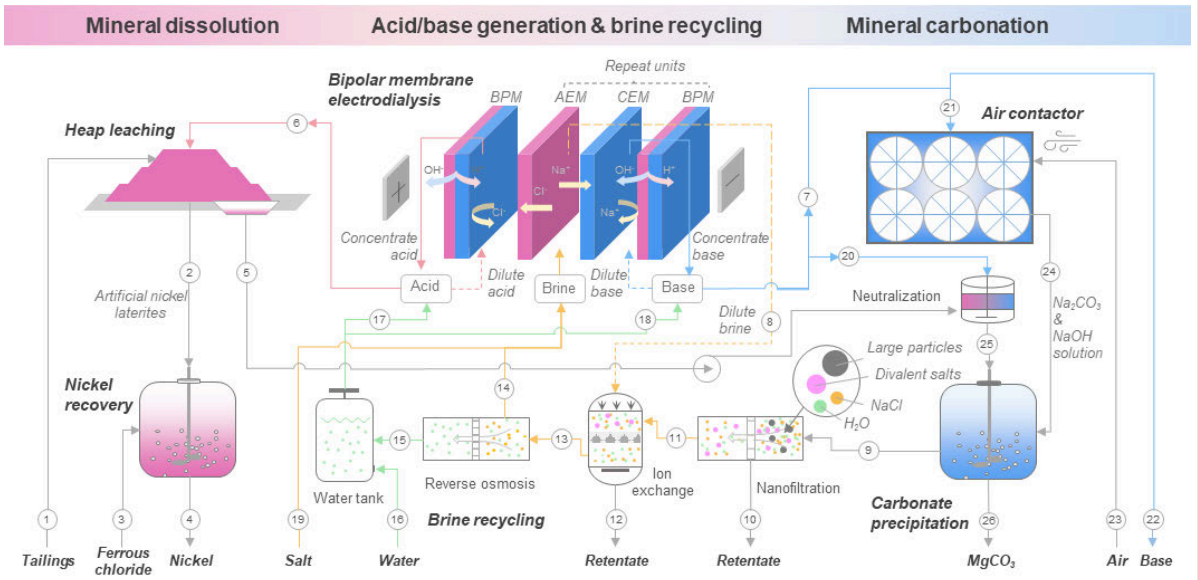


Figure 1. Process flow diagram for critical metal recovery and carbon mineralization from mine tailings (reproduced from Su et al., in review). Note: the process can use either sulfuric (preferred) or hydrochloric acid.

The Opportunity for Advancement

We have demonstrated that the components of our process (**Figure 1**) work at the laboratory bench scale. Magnesium is released into solution while Ni and Co are found in both solution and newly precipitated Fe-(oxy)hydroxides following silicate mineral dissolution. The efficiency of Ni and Co recovery (~80%) from Fe-(oxy)hydroxides is high (**Figure 2**) and we have shown up to 89% magnesium carbonation efficiency from brines in collaboration with an industrial collaborator (Turvey et al., in review). We have recently demonstrated the feasibility of the leaching process at the 1-tonne scale. However, our laboratory and demonstration results show that total recoveries of both Mg and Ni from ultramafic mine tailings remain lower than required to make the process economical: 10–22% using 0.2 M and 15–52% at 0.8 M sulfuric acid using daily acid treatments for 30 days. These results indicate that substantial untapped potential to recover Mg, Ni and Co remains in the ultramafic residues under the leaching conditions tested. **Extracting the remaining Mg for carbonation and Ni + Co for recovery through optimized heap leach design is the central opportunity for advancement of this technology.**

While the core process works, **heap leach optimization for Mg and Ni + Co recovery is the single greatest bottleneck to increasing CO_2 mineralization efficiency and improving process economics.** Ultramafic rocks are seldom heap leached, as such there is no plug-and-play solution that currently exists. With support from Frontier, we will design and test a novel, optimized heap leach technology that could transform abundant mine tailings into a dual-purpose resource for durable CO_2 storage and critical metal supply, which supports the clean energy transition.

Our proposed work has potential for **broad impact on the field of CDR** by developing a novel heap leaching process for the simultaneous treatment of kilotonne- or megatonne-scale piles of ultramafic tailings at many mines worldwide. Heap leaching is an existing and scalable technology for metal recovery with regulatory and permitting frameworks already in place in many countries. If successful, our design of an optimized heap leaching process would unlock the abundant Mg, Ni and Co stored in silicate minerals, providing an order of magnitude increase in CDR potential by embedding metal resource extraction into enhanced surficial weathering. Given current estimates for increased demand for Ni, Co and other metals found in ultramafic rocks, our process could potentially be scaled to remove ~1 Gt CO_2/year by 2100 (Wilson & Hamilton, 2022).

References

Hamilton, J.L., Wilson, S., Morgan, B., Harrison, A.L., Turvey, C.C., Paterson, D.J., Dipple, G.M., and Southam, G., 2020, Accelerating mineral carbonation in ultramafic mine tailings via direct CO_2 reaction and heap leaching with potential for base metal enrichment and recovery: Economic Geology, 115, 303–323.

Power, I.M., Dipple, G.M., and Southam, G., 2010, Bioleaching of ultramafic tailings by *Acidithiobacillus* spp. for CO_2 sequestration: Environmental Science & Technology, 44, 456–462.

Su, D., McBride, A.L., Pereira, R.J.L., Yaddala, J., Turvey, C.C., Hussain, A., Moyo, A., Hamilton, J.L., Wang, Z., Mann, M., Hu, J., Lammers, L., Wilson, S., Frierdich, A.J., Van der Spek, M., and Renforth, P., in review, Net negative carbon dioxide nickel mining examined through prospective technoeconomic assessment: Natural Communications.

Turvey, C.C., Hussain, A., Majusiak, A., Su, D., Mann, M., Wang, Z., Moyo, A., Kronholm, B., Yaddala, J., Rausis, K., Power, I.M., Spence, J., McBride, A., Hamilton, J.L., Lammers, L.N., Renforth, P., Frierdich, A., and Wilson. S., in review, Acid leaching for carbon sequestration with prospects for critical metal recovery: Environmental Science & Technology.

Wilson, S., and Hamilton, J.L., 2022, Fizzy ore processing sequesters CO_2 while supplying critical metals: PNAS: 119, e2212424119.

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

Project Goals, Location and Scale

We propose an R&D project to design and test a hybrid chemical–biological heap leaching system for CDR and critical metal recovery that can be applied to the gigatonnes of global ultramafic mine tailings, including asbestos-bearing materials at Thetford, Québec and Baie Vert, Newfoundland and Labrador, Canada. We are already in conversation with Baie Minerals leadership about a research collaboration. The proposed research could be done at either location.

Most of the project will be conducted using laboratory experiments, reactive-transport modelling and an in-depth technology assessment, culminating in a small-scale field test of the optimized heap leaching system in the final six months of the project. Within a 24-month timeline, this research will directly address uncertainties in acid consumption, reaction extent, and heap performance, leading to a modular, generally applicable and scalable system that couples durable CDR with critical metal recovery.

Research Team

This project is jointly led by Prof. Sasha Wilson (University of Alberta), Prof. Wenying Liu (The University of British Columbia), Prof. Ian Power (Trent University), Prof. Andrew Frierdich (Monash University), Dr. Jessica Hamilton (Australian Synchrotron, ANSTO) and Prof. Phil Renforth (Heriot-Watt University). Wilson is the primary administrative contact. Our groups have a strong track-record of collaboration, reflected in multiple co-authored publications from joint research projects. We bring together complementary expertise in surficial CO_2 mineralization, microbially-enhanced dissolution and carbonation kinetics, geometallurgical characterization for CDR, heap leach design and testing, and techno-economic analysis (TEA) and life cycle assessment (LCA). We have the full range of complementary engineering and applied science expertise to successfully execute the proposed project.

Key Research Questions

This project is designed to answer the following key research questions:

(1) What physical and chemical parameters lead to optimal leaching of Mg, Ni and Co from ultramafic mine tailings?
(2) Can bioleaching reduce the dependence on reagent-grade sulfuric acid while maintaining or improving Mg, Ni and Co recovery efficiencies?
(3) How does the optimized heap leach design impact critical metal recovery and downstream carbonation performance?
(4) Is the optimized method for Mg, Ni and Co recovery economically viable and does it have a net environmental benefit?

Approaches and Methods

Materials characterization and laboratory experiments are required to optimize physical properties and leaching conditions. This phase will be done iteratively, in parallel with heap leach modelling, TEA and LCA modelling. Once models and experiments have yielded an optimized design, we will use an instrumented field-scale test heap (or a metre-scale bulk container) to evaluate the design under realistic conditions.

Geometallurgical Model for CDR and Metal Extraction

Once materials have been received from either Thetford or Baie Vert, we will comprehensively characterize their mineralogical and physico-chemical properties to understand the distribution, deportment and recoverability of Mg, Ni and Co. Particle size distributions and surface areas will be

measured for input into models. Quantitative X-ray diffraction (XRD) will be used to obtain mineral abundances from representative samples and electron probe microanalysis (EPMA) will enable us to produce a mass-balance model for Mg, Ni and Co deportment in each mineral within the tailings. Synchrotron X-ray absorption spectroscopy (XAS) will be used for Ni and Co speciation, including determination of oxidation states and co-ordination chemistry, to inform their amenability to recovery via acid leaching. We have previously used this approach to estimate CDR potential and to improve targeting of first-row transition metal recovery from mine tailings (e.g., Hamilton et al., 2020; Wang et al., 2025; Honda-McNeil et al., 2026).

Optimizing Bioleaching and Comparison to Abiological Acid Leaching

We will test whether microbially mediated sulfur oxidation has the potential to sustain field-scale Mg and critical metal extraction from ultramafic tailings while minimizing external acid demand. Laboratory trials will use materials from Thetford or Baie Vert that are amended with waste sulfur. Our approach is differentiated from those of others by our previous experience with bioleaching of ultramafic tailings for CDR. Laboratory column experiments will follow a similar design to that used by Power et al. (2010), using ultramafic tailings layered with sulfur-bearing substrates inoculated with *Acidithiobacillus* spp. Following leaching treatments, monitoring will be done for pH, sulfate production, Mg and critical metal release, and microbial activity. Results will be used to quantify bioleaching kinetics and identify optimal conditions for tailings leaching under controlled conditions, which will be compared to abiotic controls. Feedback from model development will be used to design and iteratively improve Mg, Ni and Co recovery from both the bioleaching and abiological leaching experiments.

Heap Leach Design Optimization

Here, we will simulate the effects of key heap leach design parameters on heap performance using the adapted HeapSim model. The parameters to be evaluated include heap geometry, solution application rate, solution chemistry (particularly acid concentration), aeration rate, and temperature. Heap performance will be assessed in terms of the kinetics and extent of Mg, Ni and Co release as well as acid consumption. HeapSim has been extensively applied to model industrial copper heap bioleaching by our team (Liu et al., 2018; Winarko et al., 2023). The model incorporates the fundamental processes occurring within a heap, including solution percolation, gas transport, reaction kinetics, and the activity of iron- and sulfur-oxidizing microorganisms. However, the reaction kinetics currently included in HeapSim were developed primarily for copper sulfide leaching. In this project, the model will be adapted to incorporate serpentinite dissolution kinetics. The adapted model will then be calibrated using the laboratory data generated in the above objective. The primary outcome of this objective will be a quantitative assessment of how key heap leach design parameters influence heap performance. This information is essential for guiding the optimal design of subsequent field trials described in the next objective.

Optimization of Ni + Co Recovery and Carbonation from a Field Trial

Following materials characterization, laboratory column leaching, and heap design modelling, a small-scale field trial will be carried out at either Thetford or Baie Vert. Test heaps will be designed

and evaluated at increasing scale, beginning with field barrel tests. These tests will use ultramafic mine tailings that have mineralogical and physical properties that are representative of the materials used in our laboratory studies. Sensors will be installed to monitor pH, temperature, redox and oxygen concentrations, which are key parameters controlling heap performance. Another key aspect to control is heap permeability, which is influenced by grain size distribution, the mineralogy of the waste material, and heap stacking. Leachate samples will be analyzed for dissolved Mg, Ni and Co, pH, sulfate, and other dissolved metal concentrations and samples of the Fe-(oxy)hydroxide minerals that precipitate from solution or within the heap will be reacted with aqueous Fe(II) using our patented metal recovery method (e.g., Wang et al., 2025). The results will be used to determine the %-recovery of Mg, Ni and Co.

Technology Assessment

In one of the first prospective techno-economic assessments for geochemical CDR, we have demonstrated that the performance of the complete system described in **Figure 1** (Su et al., in review) is dominated by the electrochemical (acid production) process. Costs may be substantially reduced from their first-of-a-kind (>$1000s/t CO_2) to <$500/t CO_2 as the technology scales, or potentially with the application of bioleaching using acid-generating mineral amendments. The work has highlighted the level of subsidy required in order to scale and drive down costs. This TEA is parameterized using an idealized reactive transport model to simulate heap leach processes and tuned on small scale experimental data. As part of the proposed project, we will take the next steps in developing this model. First, we will validate the TEA using the field experimental results from this project. Second, we will expand our technology assessment with a full prospective LCA.

Best in Class Approach

We have deep experience with assessing and scaling enhanced surficial weathering for CDR. Our team has previously completed multi-tonne, mid-TRL demonstrations of acid heap leaching and enhanced passive weathering of ultramafic materials, demonstrating accelerated CO_2 mineralization and co-recovery of Ni and Co, work that underpins this proposal. Importantly, we are among only a few groups globally working at the intersection of CDR and critical metal recovery, having completed several of the first studies on this topic (e.g., Hamilton et al., 2020). We have the experience and the laboratory and modelling facilities needed to succeed in this project.

References Not Previously Listed

Honda-McNeil, M.J., Wilson, S., and Locock, A.J., 2026, Quantitative analysis of critical metal distribution and recoverability from serpentinite skarn tailings from Lord Brassey mine, Tasmania: American Mineralogist.

Liu, W., Granata, G., 2018, Temperature control in copper heap bioleaching: Hydrometallurgy, 176, 26–32.

Wang, Z., Mann, M., Hamilton, J.L., Turvey, C.C., Hussain, A., Wilson, S., Su, D., McBride, A.L., Renforth, P., Lammers, L.N., Moyo, A., Yaddala, J., and Frierdich, A.J., 2025, Artificial laterite from acid leaching of ultramafic rocks: Mobilization, enrichment, and extraction of critical metals: Environmental Science & Technology, 59, 20829–20841.

Winarko, R., Dreisinger, D.B., Miura, A., Fukano, Y., Liu, W., 2023, Modelling the scale-up of the iodine-assisted chalcopyrite leaching in ferric sulfate media from reactors to columns and simulating heap operations: Hydrometallurgy, 222, 106203.

d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

Scientific and Technical

The primary scientific and technical uncertainties relate to how heap leach performance will scale with increasing size. It is unclear whether the efficiencies observed at the laboratory scale will persist at the field scale, where fluid flow, passivation, and formation of secondary minerals may limit reactions. For instance, heap permeability is a key factor controlling heap performance. Various approaches can be used to mitigate this issue, such as using lower heap height, pulsed irrigation, or mixing fine and coarse materials. These are low-to-moderate risks associated with the field trial that will be mitigated by the design of the study, which will carefully assess these possibilities during the laboratory and modelling stage.

The laboratory and modelling aspects of the study are low risk and guaranteed to deliver valuable information for further development of enhanced surficial weathering at Thetford, Baie Vert and other ultramafic mines. The full TEA and LCA models will have broad utility for other similar geochemical CDR technologies.

Project Execution

Project execution risk is low for the laboratory and modelling aspects of the study as we already possess the expertise, facilities and equipment required for those aspects of the project. Risks associated with the small-scale field trial are moderate as they depend on an industry partner providing resources, including employee time and space at the mine, to conduct the experiments. We have previously worked with government and industry partners, including several major mining companies, to conduct on-site technology demonstrations at the multi-tonne scale. Our experience with this process and our existing connections with Baie Minerals, and Frontier's with Thetford, will serve to mitigate this risk.

MRV

The risk associated with MRV is negligible at the scale on which we are working. We have developed the methods required to conduct detailed, accurate and precise mass balance calculations for recovery of Mg, Ni and Co from ultramafic ores and mine tailings using acid leaching (e.g., Power et al., 2010; Hamilton et al., 2020; Wang et al., 2025; Turvey et al., in review) and for measuring CO_2 mineralization using this process (Turvey et al., in review).

Ecosystem

The ecosystem risks are very low for the laboratory and modelling portion of this study. We have existing safety protocols for working with, and disposing of, materials containing chrysotile asbestos. These procedures have been co-developed and vetted by an industrial hygienist at the University of Alberta who is an expert in asbestos abatement. There is a low risk of acid spilling during the on-site field trial, which will be mitigated by either conducting the trial within sealed containers or pads with carefully controlled inflows and outflows.

Financial

The financial risk is low for the laboratory and modelling aspects of the project and low-to-moderate for the field-based trial. Depending on the resources available at the mine site, unexpected costs may arise. We will mitigate this risk by working closely with the team at the prospective site over the course of the project and by re-using equipment and sensors from past field trials.

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	1. A geometallurgical model for enhanced surficial weathering, carbonation and metal recovery for one asbestos mine. 2. A generally applicable, optimized acid heap leaching method (abiological and/or bioleaching) for CDR and metal recovery using serpentinite. 3. Yield and kinetic data for optimized leaching of Mg, Ni and Co from serpentinite and for carbonation of Mg-rich leachates. 4. A heap leach model for design and optimization of heap leaching using serpentinite. 5. A full prospective LCA and TEA for the CDR + Ni and Co recovery process using the optimized heap leaching results.
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	The key impacts of this project will be: 1. Determining how much of the CDR potential of ultramafic mine tailings can be accessed using acid heap leaching.

	2. Assessing whether efficiencies can be found using bioleaching. 3. The first full LCA and TEA results for this potentially highly scalable geochemical CDR technology, which could then serve as a model for other related technologies.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	**Project duration: 24 months** **Months 1–3:** Acquisition of ultramafic mine tailings from the industry partner followed by detailed mineralogical, geochemical and physico-chemical characterization. **Months 4–18:** Conduct and optimization of (bio)leaching column tests, Fe(II)-mediated critical metal recovery, heap leach model development and TEA/LCA modelling. These elements of the project will be conducted in parallel to make iterative changes to forward the goal of optimization. Analytical work will be ongoing to monitor the results of the laboratory experiments and to provide inputs for models. **Month ~19:** A field trial of the optimized heap leaching method will be conducted at a mine. **Months 20–24:** Analysis of data from the field trial and integration of data into revised leaching and technology assessment models.
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	**$885,000**